Prof. Dr. Britta Gehrke
Freie Universität Berlin

Dr. Nicolas Syrichas
Freie Universität Berlin

Prof. Dr. Alexander Kriwoluzky
Freie Universität Berlin

Dr. Florian von Muschwitz
WZB Berlin

Advanced Macroeconomic Analysis I

Winter 2026/27

Time and place

Wednesdays, 10-12 am & Thursdays, 10-12 am

First part (starting October 14th): Wednesdays FU Berlin, Garystr. 21, HS 106 & Thursdays, Garystr. 21, HS 108

Second part (starting December 9th) : Elinor-Ostrom room DIW Berlin

Summary

The course "Advanced Macroeconomic Analysis I" is the first semester of the full-year 1st year PhD sequence in macroeconomics. It serves as an introduction to methods and models of macroeconomics and to train students in basic dynamic concepts and tools used by researchers in macroeconomics, microeconomics, and econometrics.

This course is open to BSoE students, qualified master's students, or others with the instructor's permission. The course presumes familiarity with macroeconomics. The course is at an advanced level and we will move rapidly through difficult material. It is not a course for those encountering macroeconomics for the first time.

Evaluation Mid-term and final exam (120 min)

Textbooks

- M. Azzimonti, P. Krusell, A. McKay, and T. Mukoyama et al. (2026). Macroeconomics. A comprehensive textbook for first-year Ph.D. courses in macroeconomics. https://phdmacrobook.org/
- Gali, J. (2015). Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications, Princeton University Press.
- Ljungqvist, L., and T. Sargent (2018). Recursive Macroeconomic Theory 4nd ed. Cambridge, USA: MIT Press.

Part 1: Introduction to Modern Macroeconomics (Britta Gehrke and Nicolas Syrichas, week 1–8)

Description

This part of the course presents standard frameworks used in modern macroeconomics. It focuses on dynamic optimization, dynamic programming and the solution of dynamic models without and with uncertainty in the context of the neoclassical growth model and with labor market frictions.

Overview (preliminary)

1. Week 1: Introduction and framework
    - Readings: AKMM Chapters 1+2, Lucas (1976), Sims (1980) Nakamura and Steinsson (2018)
2. Week 2: Consumption and saving: Dynamic optimization, general equilibrium, recursive problems, value function
    - Readings: AKMM Chapters 4+5, LS Chapters 1+3
3. Week 3+4: The neoclassical growth model: Solving dynamic models, steady state, phase diagram, dynamic effects of shocks, planner vs decentralized equilibrium, solving linearized models, value function iteration
    - Readings: AKMM Chapters 4-6 + 10, Campbell (1994), Blanchard and Kahn (1980a)
4. Week 4+5: Frictional labor markets: Uncertainty, Markov processes, expectations, impulse response functions, aggregate vs idiosyncratic risk
    - Readings: AKMM Chapters 7, 10+20, Pissarides, C. (2000). *Equilibrium Unemployment Theory*, 2nd ed. MIT Press.
5. Week 6: Overlapping generations: equilibrium, efficiency, and policy (Nicolas Syrichas)

    Two-period lives; exchange and production economies; competitive equilibrium and capital accumulation; the golden rule and dynamic inefficiency; pay-as-you-go pensions and public debt.

    - Required readings: AKMM Chapters 5–6 (OLG sections); Krueger, Sections 8.1 and 8.3 (selected material).

- Further reading: Krueger, Section 8.2 (Ricardian equivalence); AKMM Chapter 15 (OLG and intergenerational policy sections).
- Krueger, D. (2012). *Macroeconomic Theory.* Lecture notes, University of Pennsylvania, January 26 version.

6. Week 7: Continuous-time macroeconomics I: optimization and growth (Nicolas Syrichas)

   From discrete to continuous time; discounting and flow budget constraints; the current-value Hamiltonian, dynamic programming and the Hamilton–Jacobi–Bellman equation.

   - Required readings: AKMM Chapter 9 (deterministic methods and growth); Krueger, Section 9.3.
   - Further readings: Wälde, Sections 3.2–3.4; Mukoyama, *Relationship between Maximum Principle and Dynamic Programming.*
   - Krueger, D. (2012). *Macroeconomic Theory.* Lecture notes, University of Pennsylvania, January 26 version.
   - Mukoyama, T. Selected supplementary notes: *DMP Model in Continuous Time* and *Relationship between Maximum Principle and Dynamic Programming.*

7. Week 8: Continuous-time macroeconomics II: uncertainty and labor markets (Nicolas Syrichas)

   Poisson processes and transition rates; Bellman equations with jumps; worker and firm values in continuous time; unemployment flows and the Beveridge curve; vacancy creation and wage bargaining.

   - Required readings: AKMM Chapter 9 (Poisson uncertainty); Mukoyama, *DMP Model in Continuous Time.*
   - Further readings: Wälde, Sections 5.3, 5.6.1, and 8.2–8.4; AKMM Chapter 20 (review of the DMP model).
   - Mukoyama, T. Selected supplementary notes: *DMP Model in Continuous Time* and *Relationship between Maximum Principle and Dynamic Programming.*
   - Wälde, K. (2020/21). *Advanced Macroeconomic Theory 1 (Part 2).* Lecture notes, GSEFM, Johannes Gutenberg University Mainz.

Part 2: Monetary & fiscal policy in a New-Keynesian model (Alexander Kriwoluzky, week 9–16)

Overview
This part of the course is built around the New Keynesian (NK) model. We open with a historical lecture motivating why money is not neutral, then develop the baseline NK model step by step: determinacy, optimal policy under discretion and commitment, fiscal policy, and a two-agent (TANK) extension. The final block asks why money has value and closes with the fiscal theory of the price level. Each lecture is paired with an exercise that implements and solves the corresponding model in `Dynare`.

Introduction (Week 1)

Lecture 1: A small sample of macro-history

- Required reading: Galí (2015) (Ch. 1); Nakamura and Steinsson (2018)
- Further readings: Friedman and Schwartz (1963); Christiano et al. (1999); Mussa (1986); Romer and Romer (2004)

Exercise 1: Introduction to MATLAB and `Dynare`

- Setting up `Dynare`, simulating a simple stochastic model, reading impulse responses: Adjemian et al. (2024)

The New Keynesian model (Weeks 2–4)

Lecture 2: The baseline New Keynesian model

- Required reading: Galí (2015) (Ch. 2–3); Woodford (2003) (Ch. 3)
- Further readings: Walsh (2017) (Ch. 8); Clarida et al. (1999)

Exercise 2: Solving the three-equation NK model in `Dynare`

- Impulse responses to a monetary policy shock; calibration against Galí's benchmark
- Extension: the small open-economy NK model: Galí (2015) (Ch. 8)

Lecture 3: Determinacy and the Taylor principle

- Required reading: Galí (2015) (Ch. 4); Woodford (2003) (Ch. 4)
- Further readings: Blanchard and Kahn (1980b); Bullard and Mitra (2002); Cochrane (2011)

Exercise 3: Determinacy regions in `Dynare`

- Blanchard–Kahn conditions; mapping the (in)determinacy region of the Taylor rule

Lecture 4: Optimal monetary policy — discretion versus commitment

- Required reading: Galí (2015) (Ch. 5); Woodford (2003) (Ch. 7)
- Further readings: Kydland and Prescott (1977); Barro and Gordon (1983); Clarida et al. (1999)

Exercise 4: Optimal policy in `Dynare`

- Ramsey policy, optimal simple rules and discretion (`ramsey_model`, `osr`, `discretionary_policy`); the stabilization bias

Fiscal policy and heterogeneity (Weeks 5 and 6)

Lecture 5: Fiscal policy in the New Keynesian model

- Required reading: Galí et al. (2007); Woodford (2011)
- Further readings: Christiano et al. (2011); Ramey (2019)

Exercise 5: Fiscal multipliers in `Dynare`

- Government-spending shocks; the size of the multiplier and the role of the monetary policy rule

Lecture 6: Heterogeneity and the two-agent NK model (TANK)

- Required reading: Bilbiie (2008); Bilbiie (2021)
- Further readings: Bilbiie (2025a,b); Kaplan and Violante (2018); Kaplan et al. (2018); Kaplan (2025)

Exercise 6: RANK versus TANK in `Dynare`

- Solving an analytical TANK model; comparing impulse responses and multipliers with RANK
- Extension: two sectors and production networks

The value of money and the price level (Weeks 7 and 8)

Lecture 7: Why money has value

- Galí (2015) (Ch. 2)

Exercise 7: Modelling money in `Dynare`

- Money-in-the-utility-function and cash-in-advance models; the optimum quantity of money

Lecture 8: The fiscal theory of the price level (FTPL)

- Required reading: Leeper (1991); Cochrane (2023) (selected chapters)
- Empirical evidence: Kliem et al. (2016); Ettmeier and Kriwoluzky (2024); Bianchi and Ilut (2017a)
- Further readings: Sims (1994); Woodford (1995); Leeper and Leith (2016); Bianchi and Ilut (2017b); Angeletos et al. (2024); Kaplan (2026)

Exercise 8: Monetary–fiscal interaction and the FTPL in `Dynare`

- Active/passive policy regimes; price-level determination under a fiscal anchor

Readings

Adjemian, S., Bastani, H., Juillard, M., Karamé, F., Maih, J., Mihoubi, F., Mutschler, W., Pfeifer, J., Ratto, M., and Villemot, S. (2024). *Dynare: Reference Manual*. CEPREMAP. Dynare Working Papers, 1.

Angeletos, G.-M., Lian, C., and Wolf, C. K. (2024). Can deficits finance themselves? *Econometrica*, 92(5):1351–1390.

Barro, R. J. and Gordon, D. B. (1983). A positive theory of monetary policy in a natural rate model. *Journal of Political Economy*, 91(4):589–610.

Bianchi, F. and Ilut, C. (2017a). Monetary/Fiscal policy mix and agents' beliefs. *Review of Economic Dynamics*, 26.

Bianchi, F. and Ilut, C. (2017b). Monetary/fiscal policy mix and agents' beliefs. *Review of Economic Dynamics*, 26:113–139.

Bilbiie, F. O. (2008). Limited asset markets participation, monetary policy and (inverted) aggregate demand logic. *Journal of Economic Theory*, 140(1):162–196.

Bilbiie, F. O. (2021). The new keynesian cross. *Journal of Monetary Economics*, 122:38–55.

Bilbiie, F. O. (2025a). Monetary policy and heterogeneity: An analytical framework. *Review of Economic Studies*, 92(4):2398–2436.

Bilbiie, F. O. (2025b). *Tractable Heterogeneous-Agent New Keynesian (THANK) Macroeconomics*. MIT Press. Forthcoming; draft chapters circulated by the author.

Blanchard, O. J. and Kahn, C. M. (1980a). The solution of linear difference models under rational expectations. *Econometrica: Journal of the Econometric Society*, pages 1305–1311.

Blanchard, O. J. and Kahn, C. M. (1980b). The solution of linear difference models under rational expectations. *Econometrica*, 48(5):1305–1311.

Bullard, J. and Mitra, K. (2002). Learning about monetary policy rules. *Journal of Monetary Economics*, 49(6):1105–1129.

Campbell, J. Y. (1994). Inspecting the mechanism: An analytical approach to the stochastic growth model. *Journal of Monetary Economics*, 33(3):463–506.

Christiano, L., Eichenbaum, M., and Rebelo, S. (2011). When is the government spending multiplier large? *Journal of Political Economy*, 119(1):78–121.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1999). Monetary policy shocks: What have we learned and to what end? In Taylor, J. B. and Woodford, M., editors, *Handbook of Macroeconomics*, volume 1, chapter 2, pages 65–148. Elsevier, Amsterdam.

Clarida, R., Galí, J., and Gertler, M. (1999). The science of monetary policy: A new keynesian perspective. *Journal of Economic Literature*, 37(4):1661–1707.

Cochrane, J. H. (2011). Determinacy and identification with taylor rules. *Journal of Political Economy*, 119(3):565–615.

Cochrane, J. H. (2023). *The Fiscal Theory of the Price Level*. Princeton University Press, Princeton, NJ.

Ettmeier, S. and Kriwoluzky, A. (2024). Active or passive? Revisiting the role of fiscal policy during high inflation. *European Economic Review*, 170.

Friedman, M. and Schwartz, A. J. (1963). *A Monetary History of the United States, 1867–1960*. Princeton University Press, Princeton, NJ.

Galí, J. (2015). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*. Princeton University Press, Princeton, NJ, 2 edition.

Galí, J., López-Salido, J. D., and Vallés, J. (2007). Understanding the effects of government spending on consumption. *Journal of the European Economic Association*, 5(1):227–270.

Kaplan, G. (2025). Fiscal-monetary interactions in the 2020's: Some insights from hank models. Technical report, University of Chicago, NBER and e61 Institute. Reserve Bank of Australia Annual Conference.

Kaplan, G. (2026). A note on equilibrium uniqueness in new keynesian models and implications for the fiscal theory of the price level. Technical report, University of Chicago, NBER and e61 Institute.

Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to hank. *American Economic Review*, 108(3):697–743.

Kaplan, G. and Violante, G. L. (2018). Microeconomic heterogeneity and macroeconomic shocks. *Journal of Economic Perspectives*, 32(3):167–194.

Kliem, M., Kriwoluzky, A., and Sarferaz, S. (2016). Monetary-fiscal policy interaction and fiscal inflation: A tale of three countries. *European Economic Review*.

Kydland, F. E. and Prescott, E. C. (1977). Rules rather than discretion: The inconsistency of optimal plans. *Journal of Political Economy*, 85(3):473–492.

Leeper, E. M. (1991). Equilibria under 'active' and 'passive' monetary and fiscal policies. *Journal of Monetary Economics*, 27(1):129–147.

Leeper, E. M. and Leith, C. (2016). Understanding inflation as a joint monetary–fiscal phenomenon. In Taylor, J. B. and Uhlig, H., editors, *Handbook of Macroeconomics*, volume 2, chapter 30, pages 2305–2415. Elsevier, Amsterdam.

Lucas, R. E. (1976). Econometric policy evaluation: A critique. *Carnegie-Rochester Conference Series on Public Policy*, 1:19–46.

Mussa, M. (1986). Nominal exchange rate regimes and the behavior of real exchange rates: Evidence and implications. *Carnegie-Rochester Conference Series on Public Policy*, 25:117–214.

Nakamura, E. and Steinsson, J. (2018). Identification in macroeconomics. *Journal of Economic Perspectives*, 32(3):59–86.

Ramey, V. A. (2019). Ten years after the financial crisis: What have we learned from the renaissance in fiscal research? *Journal of Economic Perspectives*, 33(2):89–114.

Romer, C. D. and Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American Economic Review*, 94(4):1055–1084.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica: Journal of the Econometric Society*, pages 1–48.

Sims, C. A. (1994). A simple model for study of the determination of the price level and the interaction of monetary and fiscal policy. *Economic Theory*, 4(3):381–399.

Walsh, C. E. (2017). *Monetary Theory and Policy*. MIT Press, Cambridge, MA, 4 edition.

Woodford, M. (1995). Price-level determinacy without control of a monetary aggregate. *Carnegie-Rochester Conference Series on Public Policy*, 43:1–46.

Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press, Princeton, NJ.

Woodford, M. (2011). Simple analytics of the government expenditure multiplier. *American Economic Journal: Macroeconomics*, 3(1):1–35.